LIFTING A GEOMETRIC POINT THROUGH A POINT OF AN ETALE MORPHISM OF ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 03FU: sources/stacks-chow/tags.txt, source lines 4505–4505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\overline{x} : \mathrm{Spec}(k) \to X$ be a geometric point of X lying over $x \in |X|$. Let $\varphi : U \to X$ be an étale morphism of algebraic spaces and let $u \in |U|$ with $\varphi(u) = x$. Then there exists a geometric point $\overline{u} : \mathrm{Spec}(k) \to U$ lying over u with $\overline{x} = \varphi \circ \overline{u}$.*

Proof. Choose an affine scheme U' with $u' \in U'$ and an étale morphism $U' \to U$ which maps u' to u. If we can prove the lemma for $(U', u') \to (X, x)$ then the lemma follows. Hence we may assume that U is a scheme, in particular that $U \to X$ is representable. Then look at the cartesian diagram

$$\begin{array}{ccc} \mathrm{Spec}(k) \times_{\overline{x},X,\varphi} U & \xrightarrow[\mathrm{pr}_2]{} & U \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle \varphi} \\ \mathrm{Spec}(k) & \xrightarrow{\overline{x}} & X \end{array}$$

The projection pr_1 is the base change of an étale morphisms so it is étale, see Lemma 03FU. Therefore, the scheme $\mathrm{Spec}(k) \times_{\overline{x},X,\varphi} U$ is a disjoint union of finite separable extensions of k, see Morphisms, Lemma 02GL. But k is algebraically

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closed, so all these extensions are trivial, so $\mathrm{Spec}(k) \times_{\overline{x}, X, \varphi} U$ is a disjoint union of copies of $\mathrm{Spec}(k)$ and each of these corresponds to a geometric point $\overline{u}$ with $\varphi \circ \overline{u} = \overline{x}$. By Lemma 03H4 the map

$$|\mathrm{Spec}(k) \times_{\overline{x}, X, \varphi} U| \longrightarrow |\mathrm{Spec}(k)| \times_{|X|} |U|$$

is surjective, hence we can pick $\overline{u}$ to lie over u. $\square$